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Do Sheaf Neural Networks Use Holonomy? A Measure–Intervene–Control Study

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Abstract

Geometric architectures make interpretable structure available to a model, but predictive accuracy alone does not show that a trained model uses that structure. We study whether holonomy—the ordered product of learned SO(2) transports around a cycle—acts as a mechanism for triangle counting in three-layer Neural Sheaf Diffusion (NSD). Our *exact-tied* model reuses one learned connection across all propagation steps, making three-edge loop compositions available to the network. We first measure triangle holonomy and then intervene on frozen checkpoints by replacing the learned connection with a synchronized cycle-flat connection, a shuffled assignment of learned rotations, or identity transports. In each case, we rebuild the sheaf Laplacian without retraining. Training produces nonflat connections, with substantially more twist at the largest training size. The three replacements have negligible mean effects with 30 or 100 training graphs but cause large increases in MSE that vary substantially across seeds with 1,000 or 3,000 graphs. This result shows that predictions depend on the learned assignment of transports to edges. The controls do not justify a stronger conclusion. Identity-trained NSD can outperform learned-connection NSD, twist predicts neither counting accuracy nor the MSE increase from flattening within training size, and random 6-regular graphs can acquire twist without counting skill. The learned connection can therefore affect predictions from a checkpoint even when nonidentity transport is neither necessary nor demonstrably beneficial, and the present interventions do not isolate holonomy as the causal counting mechanism.

Keywords: geometric deep learning, sheaf neural networks, cellular sheaves, holonomy, interpretability

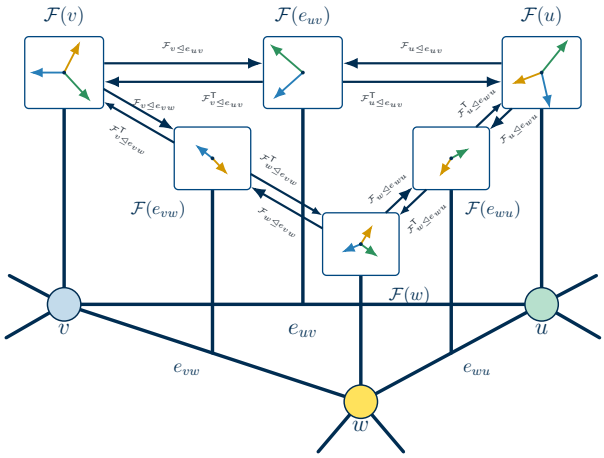


Figure 1: **A cellular sheaf over a triangle.** Restriction maps carry signals between the vector spaces attached to vertices and edges. Comparing the two endpoint maps induces transport between neighboring vertices. Colors track the source vertex.

1. Introduction

Geometric architectures encode priors about symmetry, topology, and transport. These priors determine which computations a model can represent, but they do not identify which computation a trained model actually uses. Three claims must therefore be separated. An architecture can make a geometric mechanism available, training can produce the corresponding geometric structure, and predictions can depend on that structure. Each claim requires different evidence.

Cellular Sheaf Neural Networks (SNNs) provide a concrete setting for this distinction. In the orthogonal-bundle family, learned edgewise transports form a discrete connection whose loop products can make propagation path dependent (Hansen and Ghrist, 2019; Hansen and Gebhart, 2020; Bodnar et al., 2022; Barbero et al., 2022). The resulting holonomy is a plausible finite-depth mechanism, but it is not unambiguously beneficial. Nonidentity holonomy also restricts the globally consistent signals preserved by diffusion. Section 2 fixes the formal notation, and Appendix B gives the layer-aware loop construction.

We study triangle counting because triangles are the shortest cycles with nontrivial holonomy and cannot generally be counted by 1-WL-bounded message-passing GNNs (Morris et al., 2019; Chen et al., 2020). To make a three-edge loop product available across depth, we use a three-layer NSD that reuses one learned connection across layers, which we call *exact-tied*. This design enables a holonomy-based computation but does not imply that training uses it.

The study follows the resulting claims in order. We first determine whether training produces nontrivial triangle holonomy. We then test whether frozen predictions depend on the learned connection. Finally, we ask whether any such dependence shows task-level value or identifies holonomy as the counting mechanism. This progression defines our measure–intervene–control analysis and prevents checkpoint sensitivity from being interpreted directly as causal evidence (Elazar et al., 2021). We make three contributions.

- We formalize, in an idealized linear setting, how holonomy changes finite-depth return signals while restricting the global-section space preserved by diffusion.
- We introduce a synchronized-flat intervention for testing whether a trained checkpoint depends on nontrivial cycle products in its learned connection.
- We empirically separate learned loop products, checkpoint reliance, and task-level value. This distinction shows that dependence on the assignment of learned transports to edges does not by itself make holonomy necessary, beneficial, or causal for triangle counting.

2. Background and Related Work

Sheaf diffusion and learned connections. A cellular sheaf $\mathcal{F}$ assigns vector spaces, or stalks, to the vertices and edges of a graph, together with a restriction map $\mathcal{F}_{v \trianglelefteq e} : \mathcal{F}(v) \to \mathcal{F}(e)$ for each incidence (Hansen and Ghrist, 2019). For an oriented edge $e = (u, v)$, the coboundary measures disagreement in the common edge stalk as $(\delta_{\mathcal{F}} x)_e = \mathcal{F}_{v \trianglelefteq e} x_v - \mathcal{F}_{u \trianglelefteq e} x_u$. This gives the unnormalized sheaf Laplacian $L_{\mathcal{F}} = \delta_{\mathcal{F}}^{\top} \delta_{\mathcal{F}}$. SNNs propagate node-stalk signals using $\Delta_{\mathcal{F}} = D^{-1/2} L_{\mathcal{F}} D^{-1/2}$, where D is the block diagonal of $L_{\mathcal{F}}$ (Hansen and Gebhart,

2020; Bodnar et al., 2022; Suk et al., 2022). The notation table at the start of the appendix collects the symbols used throughout the paper.

In the orthogonal bundle family, all stalks have rank d and the restriction maps are orthogonal. On an edge $e = \{u, v\}$, the two restrictions induce the transport $T_{v \leftarrow u} = (\mathcal{F}_{v \trianglelefteq e})^\top \mathcal{F}_{u \trianglelefteq e}$, with $T_{u \leftarrow v} = T_{v \leftarrow u}^{-1}$. These transports form a discrete $O(d)$ connection (Barbero et al., 2022). Our models use $\mathrm{SO}(2)$, so each transport is a planar rotation $R(\alpha)$ and the identity transport is I_2. Figure 1 illustrates the stalks and restriction maps.

Holonomy and representation geometry. For an oriented closed path, *holonomy* is the ordered product of its edge transports. It compares a vector before and after transport around the loop, so a nonidentity product records path dependence. For an $\mathrm{SO}(2)$ connection, its rotation angle is independent of the local frames used to describe the stalks (Singer and Wu, 2012). Sevetlidis and Pavlidis (2026) use holonomy to diagnose learned representations from locally estimated transports, whereas we measure transports learned directly by a sheaf model. Appendix B.1 specifies the realized, layer-aware products.

Connection reliance and alternative solution paths. *Task-level value* asks whether learned transports outperform a separately optimized Identity Sheaf Network (ISN). *Checkpoint reliance* asks whether altering the connection inside an already-trained model increases its prediction error. Prior work distinguishes these questions by comparing learned-connection models with ISNs (Hernandez Caralt et al., 2026; Liu, 2026). We extend this distinction to cycle products. Our Simplicial Complex Convolutional Network (SCCN) positive control receives the clique complex, where each target triangle is supplied directly as a 2-simplex (Hajij et al., 2025). Appendix C.3 specifies all control inputs and training configurations.

3. Why Holonomy Might Help and Hurt

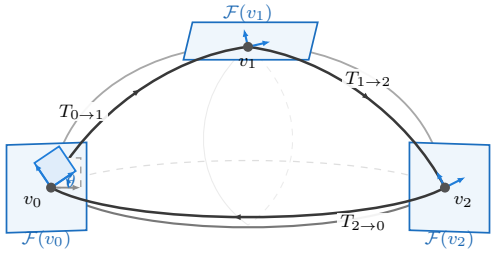


Figure 2: **Finite-depth loop sensitivity on a triangle.** Transport around the triangle returns to v_0 with a visible frame mismatch.

Sheaf diffusion and finite-depth transport emphasize different properties of the same connection. To expose the distinction, consider an idealized linear version of exact-tied propagation. Let $X^{(\ell)}$ stack the node-stalk signals after step ℓ, and fix $\mathcal{F}$ and a diffusion step size $\eta > 0$. After L steps, $X^{(L)} = (I - \eta\Delta_{\mathcal{F}})^L X^{(0)}$. In the stable limit,

$$\lim_{L\to\infty} X^{(L)} = \Pi_{\ker(\Delta_{\mathcal{F}})} X^{(0)}. \tag{1}$$

This limit holds when $0 < \eta < 2/\lambda_{\max}(\Delta_{\mathcal{F}})$, where $\lambda_{\max}$ is the largest eigenvalue and $\Pi_{\ker \Delta_{\mathcal{F}}}$ projects onto the kernel of the Laplacian. For the unnormalized Laplacian, $\ker(L_{\mathcal{F}}) \cong H^0(G;\mathcal{F})$, the space of global sections that remain consistent across every edge. Degree normalization rescales this kernel, so $\ker(\Delta_{\mathcal{F}})$ is isomorphic to the same space. Repeated diffusion preserves these globally consistent components and damps the rest (Bodnar et al., 2022, Section 3).

At finite depth, the iteration has not yet reached this limiting projection, so messages arriving along different paths can still disagree. For an oriented loop $C = (v_0, v_1, \ldots, v_{k-1}, v_0)$ that starts and ends at the base node v_0, its holonomy is

$$H_C = T_{v_0 \leftarrow v_{k-1}} \cdots T_{v_2 \leftarrow v_1} T_{v_1 \leftarrow v_0}. \tag{2}$$

A connection is *flat* when $H_C = I_2$ for every loop C. If $H_C \neq I_2$, a vector transported around the loop need not return unchanged. Proposition 1 makes this path disagreement explicit for the shortest possible cycle, a triangle (Singer and Wu, 2012).

Proposition 1 Finite-depth loop sensitivity. *Consider an isolated triangle with a fixed* SO(2) *connection, and let S be its connection-adjacency operator. For a node v and either oriented triangle traversal C based at v, if $H_C = R(\theta_C)$, then*

$$(S^3)_{vv} = H_C + H_C^{-1} = 2\cos(\theta_C) I_2.$$

In particular, a flat connection gives $(S^3)_{vv} = 2I_2$.

Repeated application of a fixed connection can therefore distinguish flat from nonflat triangle transport. This proposition identifies an idealized linear signal, not the exact computation performed by NSD. The full architecture also includes Laplacian normalization, learned channel transformations, residual terms, and nonlinearities.

The same path disagreement has the opposite effect on global consistency. A global section must return unchanged after transport around every loop, so nonidentity holonomy restricts the vectors that diffusion can preserve (Bodnar et al., 2022, Proposition 4 and Lemma 6).

Proposition 2 Holonomy obstruction to global sections. *Let $\mathcal{F}$ be a connected discrete* SO(2) *bundle, let $\Gamma(\mathcal{F})$ denote its global-section space, and fix a base node v. Then*

$$\Gamma(\mathcal{F}) \cong \{x_v \in \mathbb{R}^2 \mid H_C x_v = x_v \text{ for every loop } C \text{ based at } v\}.$$

Consequently, a flat connection has a two-dimensional global-section space, whereas one nonidentity loop holonomy makes this space trivial.

Together, the propositions formalize why holonomy might help and hurt. It changes finite-depth return signals while reducing the globally consistent subspace preserved by diffusion (Bodnar et al., 2022). They are idealized statements about a fixed linear connection, not claims about the computation performed by NSD. Exact-tied NSD probes whether the full nonlinear model exploits the repeated-connection signal, and our checkpoint interventions test whether trained predictions depend on the resulting loop products. Proofs are given in Appendix B.3.

4. Methodology

We measure H_C, evaluate frozen-checkpoint connection replacements, and compare alternative predictors. Appendices B–D give the detailed measurement, aggregation, experimental, and intervention protocols.

4.1. Measuring Triangle Holonomy

For each input graph, we extract the realized exact-tied connection and compute H_C for every oriented triangle $C = (v_0, v_1, v_2, v_0)$ using Eq. 2. This operator-level readout does not select a hidden activation. Appendix B.1 records the layer-aware construction and the conventions used for nonorthogonal restriction families.

For the main SO(2) experiments, the triangle-loop product H_C is a planar rotation. We measure its rotation magnitude by

$$\text{twist}(H_C) = \arccos\left(\frac{\text{tr}(H_C)}{2}\right). \tag{3}$$

Twist is the angle by which a frame fails to return to its starting orientation, and it is independent of an arbitrary orthonormal choice of stalk coordinates. Appendix B.1 defines gain and flip for diagonal and general restriction families. Lemma 3 in Appendix B.3 proves basis independence.

Our primary summary is the triangle-weighted mean of $\text{twist}(H_C)$ over the test set. Every triangle contributes equally, so graphs with more triangles contribute more observations. When we discuss *twist change*, we mean the trained value minus the value at initialization for the same model and evaluation set. Equation (12) in Appendix B.2 gives the aggregation formula. Appendix D.4 reports numerical checks of the measurement and interventions.

4.2. Connection Interventions

To test whether predictions depend on the learned connection, we replace its transports while keeping every other checkpoint parameter fixed. Let E^+ contain one chosen orientation of each undirected edge. For an oriented edge $e = (u \leftarrow v) \in E^+$, write the learned transport as $T_e = R(\alpha_e)$. The reverse orientation $\bar{e}$ uses $T_{\bar{e}} = T_e^{-1}$. For Shuffled, $\pi : E^+ \to E^+$ denotes a random permutation of these stored orientations. Table 1 defines the three replacements.

Table 1: **Frozen-checkpoint connection interventions.** The table gives each replacement transport and its effect on connection structure.

Intervention	Intervened Transport	Effect
Synchronized Flat	$T^{\text{flat}}_{u\leftarrow v} = R(\phi_u - \phi_v)$, with synchronized node phases ϕ.	Replaces the learned connection by a data-dependent pure-gauge connection with identity holonomy on every cycle.
Shuffled	$T_e^{\text{shuffle}} = R(\alpha_{\pi(e)})$.	Preserves the learned rotation multiset but changes its assignment to edges.
Identity	$T_e^{\text{id}} = I_2$ for every edge.	Replaces the learned connection by the trivial connection.

Synchronized Flat is the only replacement that requires graph-wide reconstruction. On each connected component, spectral angular synchronization estimates ϕ from the learned edge rotations (Singer, 2011; Bandeira et al., 2013), after which Table 1 defines T^{flat}. Appendix D.1 gives the Hermitian phase-adjacency construction, normalized eigenproblem, Algorithm 1, and exact-flatness argument. Appendix D.2 fixes the Shuffled and Identity conventions, Appendix H validates the reconstruction, and Appendix G discusses its causal scope.

5. Experimental Setup

We specify the graph distributions, model configuration, evaluation sets, and controls. Appendix C gives the complete design.

Graph distributions and tasks. Let $\mathcal{D}_{\text{GU}}^{\text{hi}}$ denote the high-homophily GraphUniverse distribution generated by a degree-corrected stochastic block model (Van Langendonck et al., 2026), and let $\mathcal{D}_{6\text{reg}} = \{G : |V(G)| = 100,\ \deg_G(v) = 6\ \forall v\}$ denote the fixed-degree control. Because synthetic triangle counts can correlate with graph size and degree statistics (Geirhos et al., 2020), we also evaluate shortcut baselines. For split s, the target is

$$z_s(G) = \frac{\log(1 + y(G)) - \mu_s}{\sigma_s},$$

where μ_s and σ_s are computed only on $\mathcal{D}_{\text{train}}^{(s)}$. Hence every reported MSE is measured in standardized log-count space. The low-homophily distribution $\mathcal{D}_{\text{GU}}^{\text{lo}}$ tests whether the twist–error result is unique to the high-homophily distribution. Table 2 and Appendix C.1 give the complete distributional parameters.

Models and optimization. We use three-layer exact-tied NSD (Bodnar et al., 2022). Sheaf models use two-dimensional stalks, hidden width 32, mean graph pooling, and a multilayer perceptron (MLP) readout. A diagonal NSD control restricts its derived edge maps to coordinatewise transformations and therefore cannot represent a continuous planar rotation. Neural Sheaf Propagation (NSP) (Suk et al., 2022) provides the architectural comparison in Appendix E.1. The NSD training-size sweep uses $N_{\text{train}} \in \{30, 100, 300, 1000, 3000\}$ and ten model seeds. Table 3 summarizes the architecture and training settings. Table 4 gives the evaluation sets and generator seeds.

Intervention evaluation and controls. The learned connection and the three replacements in Table 1 are evaluated on the same 200-graph test set $\mathcal{D}_{\text{test},200}$ for each checkpoint. If $\mathcal{L}_r$ is the test MSE under replacement r, we define $\Delta_r = \mathcal{L}_r - \mathcal{L}_{\text{learned}}$. Thus $\Delta_r > 0$ is an increase in MSE relative to the learned connection. Appendices C.2 and D give the evaluation and per-graph reconstruction details, including rebuilding $\Delta_{\mathcal{F}}$ with all non-connection parameters fixed.

The ISN and SCCN, together with training-mean ($z_s = 0$) and degree-ridge controls, test alternative predictors that can achieve low error. For comparisons across graph distributions, $\Delta_{\text{const}} = \mathcal{L}_{\text{model}} - \mathcal{L}_{\text{mean}}$, so negative values outperform the training-mean baseline. Appendix C.3 gives the complete control definitions.

6. Results

We first quantify the learned triangle twist, then test its effect on predictions and whether the result is unique to triangle counting.

6.1. Training produces nontrivial triangle holonomy

Across the ten-seed exact-tied NSD sweep, mean final triangle twist rises from 0.0201 ± 0.0083 rad at $N_{\text{train}} = 30$ to 0.7804 ± 0.3178 rad at $N_{\text{train}} = 3000$ (Figure 8a). Training therefore produces substantial triangle twist. This result shows that nontrivial loop products are present, but not whether predictions use them.

6.2. Connection replacements increase MSE after training on more graphs

On $\mathcal{D}_{\text{test},200}$, Synchronized Flat, Shuffled, and Identity have negligible mean effects at $N_{\text{train}} \in \{30, 100\}$ but increase MSE as the training set grows (Figure 3b). At $N_{\text{train}} = 1000$, the mean MSE increases are $\Delta_{\text{Flat}} = 1.532$, $\Delta_{\text{Shuffled}} = 1.668$, and $\Delta_{\text{Identity}} = 1.956$. The increases remain large and vary substantially across seeds at $N_{\text{train}} = 3000$, with no stable ordering among replacements. The checkpoints therefore rely on the learned assignment of T_e to edges, but the broad changes made by Synchronized Flat to both T_e and $\Delta_{\mathcal{F}}$ prevent Δ_{Flat} from isolating holonomy.

This checkpoint reliance is distinct from task-level value. The ISN can outperform learned-connection NSD at moderate and large training sizes, reaching 0.179 ± 0.117 MSE at $N_{\text{train}} = 3000$ over ten seeds (Figure 3a), even though post-training Identity increases the MSE of learned checkpoints. Appendix E gives the ten-seed sweep and inference. Appendix D.3 reports the accompanying transport and loop changes, and Appendix F gives the complete training-time controls.

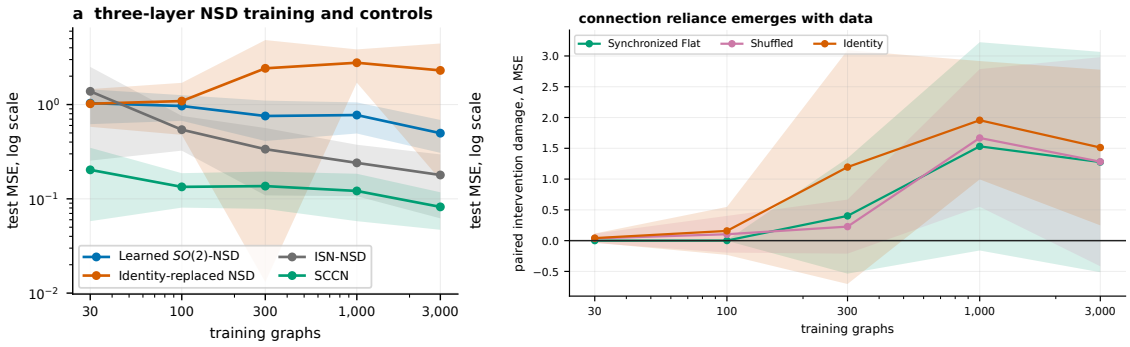


Figure 3: **Task-level value and checkpoint reliance diverge.** (a) MSE on $\mathcal{D}^{(s)}_{\text{test},8}$ for learned SO(2)–NSD and the training-mean, diagonal NSD, ISN, degree-ridge, and SCCN controls. (b) The increase in MSE under each intervention on $\mathcal{D}_{\text{test},200}$. Lines and bands show the mean and sample standard deviation over ten checkpoints.

6.3. Twist does not explain counting performance

Across the 20 exact-tied NSD checkpoints from the high- and low-homophily counting datasets, twist change is not negatively associated with Δ_{const} (Pearson $r = 0.33$, $p = 0.16$, Figure 4a). The positive sign is opposite to the association expected if greater twist improved counting. Twist also does not predict the MSE increase under Synchronized Flat after centering within training size (Pearson $r = -0.190$, $p = 0.185$, Spearman $\rho = 0.077$, $p = 0.593$).

The task also admits alternative predictors. Random 6-regular graphs can acquire twist without counting skill (Figure 4b). Community detection produces larger mean twist changes than triangle counting for both NSP and NSD at $N_{\text{train}} = 30$ (Appendix E.3 and Figure 9). Conversely, at $N_{\text{train}} = 30$, SCCN reaches 0.2030 ± 0.1451 MSE on $\mathcal{D}^{(s)}_{\text{test},8}$ and 0.2515 ± 0.1427 on $\mathcal{D}_{\text{test},200}$ over ten seeds, while degree ridge reaches 0.053 ± 0.028 MSE. Together with the ISN, these controls show that low counting error does not require the learned nonidentity connection of a particular checkpoint. Appendices E and F give the complete task-comparison and shortcut analyses.

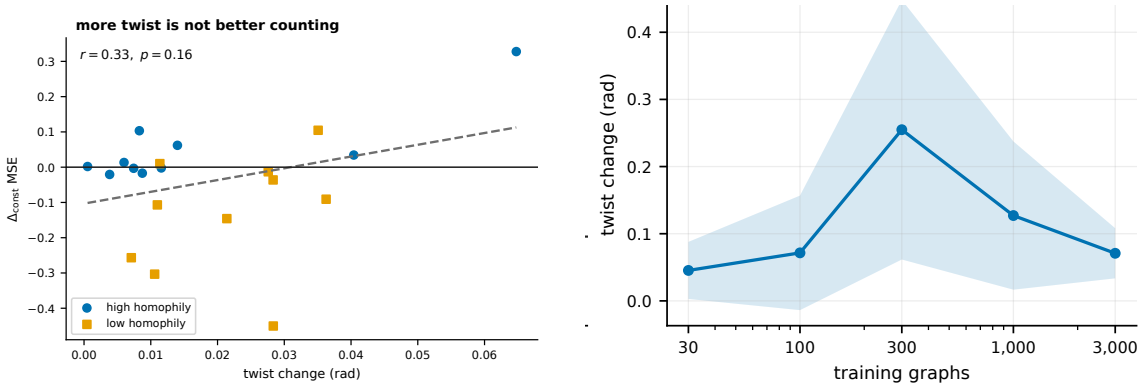


Figure 4: **Twist changes without improved counting.** (a) Larger twist change is not associated with lower error relative to the training-mean predictor. (b) On random 6-regular graphs, twist changes across training sizes while diagonal and SO(2) NSD remain near the same baseline.

Training produces nontrivial triangle holonomy. Frozen predictions become sensitive to the learned assignment of transports to edges after training on more graphs. However, twist predicts neither counting performance nor the MSE increase under flattening, and the controls identify alternative predictors with low error.

7. Discussion and Conclusion

Taken together, the experiments show that a learned connection can affect predictions from a checkpoint. The three replacements show co-adaptation to the assignment of transports to edges, not reliance on any one measured transport or loop property.

These results do not identify holonomy as the causal counting mechanism. Synchronized Flat enforces $H_C = I_2$ but also changes T_e, $\Delta_{\mathcal{F}}$, and later-layer representations. Twist predicts neither counting performance nor the MSE increase under flattening, and fixed-degree graphs can develop twist without skill. The ISN, SCCN, and degree ridge further show that checkpoint reliance implies neither task-level value nor necessity.

The scope is limited to synthetic graphs, three-layer exact-tied NSD, rank-two SO(2), and triangle loops. The main trained-model comparisons use ten seeds, while the appendix reports one descriptive three-seed NSP fixed-degree size sweep (Appendix G). Geometric structure should be measured, intervened on, and compared against alternative predictors before being treated as a predictive mechanism.

8. Future Work

Future work should determine what relational information learned connections encode beyond the alignment of stalk coordinates. The analysis should then extend from triangles to longer cycles and use sufficient propagation depth to expose their loop products. Finally, activation tracing and localized loop interventions could identify which cycles alter intermediate signals and individual predictions. These experiments could characterize the obstructions induced by learned cycle products and connect learned transport structure to a concrete computation performed by the network.

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Notation and Abbreviations

The table distinguishes similarly named objects used in the sheaf geometry, synchronization, and evaluation parts of the paper.

Symbol	Meaning
Sheaf operators and transport	
$\mathcal{F}(v), \mathcal{F}(e)$	Vertex and edge stalks.
$\mathcal{F}_{v \trianglelefteq e}$	Restriction from vertex v to incident edge e.
$\delta_{\mathcal{F}}$	Sheaf coboundary operator.
$L_{\mathcal{F}}$	Unnormalized Laplacian $\delta_{\mathcal{F}}^{\top}\delta_{\mathcal{F}}$.
D	Block diagonal of $L_{\mathcal{F}}$.
$\Delta_{\mathcal{F}}$	Normalized sheaf Laplacian $D^{-1/2}L_{\mathcal{F}}D^{-1/2}$.
$T_{v \leftarrow u}$	Directed transport from u to v.
S	Connection-adjacency operator.
$\Gamma(\mathcal{F})$	Global-section space.
Cycles and loop readouts	
$R(\alpha)$	Planar rotation through angle α.
$C,\ H_C$	Oriented cycle and its ordered loop product.
ℓ	Propagation-layer index.
twist	Rotation magnitude of a loop product.
gain, flip	Absolute determinant and determinant sign.
$\mathcal{T}(G)$	Undirected triangles in graph G.
$\mathcal{D}, \overline{\theta}$	Evaluation set and triangle-weighted mean twist.
Model abbreviations	
L, d, f	Propagation depth, stalk dimension, and hidden feature width.
$X^{(\ell)}$	Node-stalk state at propagation layer ℓ.
Θ	Trainable model parameters.
$\mathcal{S}$	Set of model seeds.
NSD / NSP	Neural Sheaf Diffusion / Propagation.
ISN	Identity Sheaf Network.
SCCN	Simplicial Complex Convolutional Network.

Symbol	Meaning
Synchronization and interventions	
E^{+}	One orientation of every undirected edge.
$z_{uv} = e^{i\alpha_{uv}}$	Complex phase of transport $v \to u$.
ϕ_u	Synchronized node angle.
$\mathcal{H}_c$	Hermitian phase adjacency on component c.
D_c	Scalar graph-degree matrix, distinct from D.
$\widehat{\mathcal{H}}_c$	Normalized phase adjacency $D_c^{-1/2}\mathcal{H}_c D_c^{-1/2}$.
q_c	Leading eigenvector of $\widehat{\mathcal{H}}_c$.
$T^{\mathrm{flat}}, T^{\mathrm{shuffle}}, T^{\mathrm{id}}$	Three replacement connections.
Data, targets, and evaluation	
$\mathcal{D}_{\mathrm{GU}}^{\mathrm{hi}}, \mathcal{D}_{\mathrm{GU}}^{\mathrm{lo}}$	High- and low-homophily GraphUniverse distributions.
$\mathcal{D}_{6\mathrm{reg}}$	Random 6-regular control distribution.
$\mathcal{D}_{\mathrm{train}}^{(s)}, \mathcal{D}_{\mathrm{val}}^{(s)}$	Training and validation sets for seed s.
$\mathcal{D}_{\mathrm{test},8}^{(s)}$	Seed-specific test set with eight graphs.
$\mathcal{D}_{\mathrm{test},200}$	Common test set with 200 graphs.
N_{train}	Number of training graphs.
$y(G)$	Raw triangle count.
s	Data/model seed index.
μ_s, σ_s	Training statistics of $\log(1+y)$.
$z_s(G)$	Standardized target, distinct from z_{uv}.
$\mathcal{L}_r$	Test MSE under replacement r.
Δ_r	MSE difference $\mathcal{L}_r - \mathcal{L}_{\mathrm{learned}}$, distinct from $\Delta_{\mathcal{F}}$.
Δ_{const}	Model MSE minus training-mean MSE.

Appendix A. Model Equations

Let $X^{(\ell)}$ denote node-stalk features, $\Delta_{\mathcal{F}}$ the normalized sheaf Laplacian, τ the NSP leapfrog step size, and $H^{(\ell)}$ the stacked vertex, edge, and triangle features used by SCCN. The

models differ as follows.

$$\text{SNN:} \quad X^{(\ell+1)} = \Phi_\ell\Big(X^{(\ell)}, \Delta_{\mathcal{F}^{(\ell)}} X^{(\ell)}\Big), \tag{4}$$

$$\text{NSD:} \quad X^{(\ell+1)} = X^{(\ell)} - \sigma\Big(\Delta_{\mathcal{F}^{(\ell)}}(I_n \otimes W_1^{(\ell)}) X^{(\ell)} W_2^{(\ell)}\Big), \tag{5}$$

$$\text{ISN:} \quad T_{v \leftarrow u} = I_2 \quad \text{in the NSD update for every edge,} \tag{6}$$

$$\text{NSP:} \quad X_{t+1} = 2X_t - X_{t-1} - \tau^2 \sigma\big(\Delta_{\mathcal{F}(t)}(I_n \otimes W_1^t) X_t W_2^t\big), \tag{7}$$

$$\text{SCCN:} \quad H^{(\ell+1)} = \psi\Big(\widetilde{D}_{\mathcal{K}}^{-1/2} \widetilde{A}_{\mathcal{K}} \widetilde{D}_{\mathcal{K}}^{-1/2} H^{(\ell)} W^{(\ell)}\Big). \tag{8}$$

Here Φ_ℓ denotes a generic SNN update. ISN is NSD with identity transport fixed during training. NSP uses the two-state leapfrog update, and the $-X_{t-1}$ and τ^2 terms must be retained. The unit-step convention sets $\tau = 1$. Exact-tied NSP uses the same $\Delta_{\mathcal{F}}$ in all three steps. Finally, $\widetilde{A}_{\mathcal{K}}$ is the clique-complex adjacency used by SCCN (Hansen and Gebhart, 2020; Bodnar et al., 2022; Suk et al., 2022; Yang et al., 2022).

Appendix B. Loop Measurement and Theoretical Foundations

This section supplies the mathematical foundations referenced from Sections 2 and 4.1. It gives the layer-aware loop construction, the test-set aggregation rule, and proofs of basis independence and the two propositions motivating the experiments.

B.1. Layer-Aware Loop Products and Readouts

Figure 5 shows the stalk-level transport measured by the loop product.

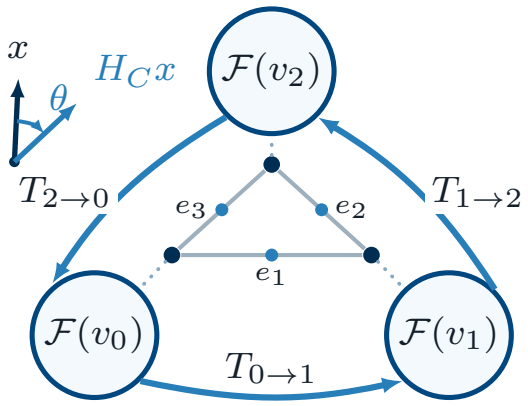


Figure 5: **Stalk-level view of triangle transport.** An arbitrary vector x in the stalk at v_0 is transported around the triangle and returns as $H_C x$. This is the stalk-level version of Figure 2.

The main text suppresses layer indices because exact tying reuses one realized connection. For completeness, let ℓ index a propagation layer and let $\mathcal{F}^{(\ell)}_{u \trianglelefteq e}$ be the restriction from the

stalk at u to the stalk at edge $e = \{u, v\}$. The corresponding directed edge map is

$$T^{(\ell)}_{v \leftarrow u} = (\mathcal{F}^{(\ell)}_{v \trianglelefteq e})^\top \mathcal{F}^{(\ell)}_{u \trianglelefteq e}, \qquad T^{(\ell)}_{u \leftarrow v} = (T^{(\ell)}_{v \leftarrow u})^{-1} \quad \text{for orthogonal restrictions.} \tag{9}$$

For a triangle $C = (v_0, v_1, v_2, v_0)$, the loop product within layer ℓ is

$$H^{(\ell)}_C = T^{(\ell)}_{v_0 \leftarrow v_2} T^{(\ell)}_{v_2 \leftarrow v_1} T^{(\ell)}_{v_1 \leftarrow v_0}. \tag{10}$$

In the exact-tied models, the same realized maps are used at every propagation step, so $H^{(\ell)}_C$ is independent of ℓ and is denoted simply by H_C in the main text. The measurement records the operator $x \mapsto H_C x$, not one observed hidden activation.

For diagonal and general restriction families, the first expression in Eq. (9) still defines a directed edge map, but it need not be invertible or isometric and the inverse relation need not hold. We therefore use the neutral terms *derived edge map* and *loop product* for those families. Strict holonomy claims concern the SO(2) models.

Additional loop readouts. For a two-dimensional loop product $H = QP$, with Q orthogonal and $P \succeq 0$ its polar decomposition, we additionally define

$$\mathrm{gain}(H) = |\det H|, \qquad \mathrm{flip}(H) = \mathrm{sgn}(\det H). \tag{11}$$

Gain is the factor by which the loop changes stalk-space area, while flip records whether it reverses orientation. For SO(2) loop products, gain is always one and flip is always $+1$. These readouts are therefore used only for the diagonal and general restriction families. For nonsingular nonorthogonal products, twist is computed from the polar factor as $\mathrm{twist}(H) = \arccos(\mathrm{tr}(Q)/2)$.

B.2. Test-Set Aggregation

Let $\mathcal{T}(G)$ be the set of undirected triangles in graph G. For test set $\mathcal{D}$ we report the triangle-weighted mean

$$\overline{\theta} = \frac{\sum_{G \in \mathcal{D}} \sum_{C \in \mathcal{T}(G)} \mathrm{twist}(H_C)}{\sum_{G \in \mathcal{D}} |\mathcal{T}(G)|}. \tag{12}$$

Each triangle contributes equally. Graphs with more triangles therefore receive more weight.

B.3. Proofs

We first prove basis independence of the loop readouts and then prove the two propositions motivating the experiments.

Lemma 3 Basis independence of the loop readouts. *The gain and flip of a loop product are unchanged by independent orthogonal changes of basis in the node stalks. If the loop product is nonsingular, its twist is unchanged as well.*

Proof Let $G_v \in O(d)$ denote an orthogonal change of basis at node v. We retain the layer index so that the statement also covers untied controls. The restrictions and directed

edge maps transform as

$$\mathcal{F}^{(\ell)}_{v\trianglelefteq e} \mapsto \mathcal{F}^{(\ell)}_{v\trianglelefteq e} G_v^\top, \qquad T^{(\ell)}_{v\leftarrow u} \mapsto G_v T^{(\ell)}_{v\leftarrow u} G_u^\top. \tag{13}$$

Substitution into the ordered loop product gives

$$(H^{(\ell)}_C)' = (G_{v_0} T^{(\ell)}_{v_0\leftarrow v_{k-1}} G^\top_{v_{k-1}}) \cdots (G_{v_1} T^{(\ell)}_{v_1\leftarrow v_0} G^\top_{v_0}) \tag{14}$$

$$= G_{v_0} H^{(\ell)}_C G^\top_{v_0}, \tag{15}$$

because every interior factor $G_v^\top G_v$ cancels. Moreover,

$$\det(GHG^\top) = \det(G)^2 \det(H) = \det(H),$$

so both $|\det H|$ and $\operatorname{sgn}(\det H)$ are unchanged. For nonsingular $H = QP$, the polar decomposition is unique. The matrix $GQG^\top$ is orthogonal, and $GPG^\top$ is positive semidefinite. The conjugated operator therefore has polar decomposition

$$GHG^\top = (GQG^\top)(GPG^\top), \tag{16}$$

so its orthogonal factor is $GQG^\top$. Trace is invariant under conjugation. In the paper's $d = 2$ setting, $\operatorname{twist}(H) = \arccos(\operatorname{tr}(Q)/2)$ is therefore unchanged. Nonsingularity is required because the orthogonal polar factor need not be unique for singular H. ■

Motivating propositions. We use the diffusion notation of Eq. 1 and the connection-adjacency operator S defined in Proposition 1. The first proof expands the three-step return block of S. The second identifies global sections with vectors fixed by every based-loop holonomy.

Proof [Proof of Proposition 1] Write the triangle vertices as $V = \{v, a, b\}$ and orient $C = (v, a, b, v)$. Block-matrix multiplication gives

$$(S^3)_{vv} = \sum_{i,j\in V} S_{vi} S_{ij} S_{jv}. \tag{17}$$

The graph has no self-loops, so a nonzero summand in Eq. (17) must correspond to a length-three closed walk from v. The isolated triangle has exactly two such walks. Therefore

$$(S^3)_{vv} = S_{vb} S_{ba} S_{av} + S_{va} S_{ab} S_{bv} \tag{18}$$

$$= T_{v\leftarrow b} T_{b\leftarrow a} T_{a\leftarrow v} + T_{v\leftarrow a} T_{a\leftarrow b} T_{b\leftarrow v} \tag{19}$$

$$= H_C + H_C^{-1}, \tag{20}$$

where inverse consistency of the edge transports identifies the second product with the holonomy of the reverse traversal. If $H_C = R(\theta_C)$, then

$$H_C + H_C^{-1} = R(\theta_C) + R(-\theta_C) = \begin{bmatrix} 2\cos\theta_C & 0 \\ 0 & 2\cos\theta_C \end{bmatrix} = 2\cos(\theta_C) I_2. \tag{21}$$

Flatness gives $H_C = I_2$, hence $(S^3)_{vv} = 2I_2$. If every propagation step uses the common scalar multiple $\widetilde{S} = cS$ for a scalar c, then $(\widetilde{S}^3)_{vv} = c^3 (S^3)_{vv}$. ■

Proof [Proof of Proposition 2] Fix a base node v. For a path p from v to a node u, write T_p for the ordered product of transports along p. If $x = (x_u)_{u \in V}$ is a global section, edge consistency gives

$$x_u = T_p x_v. \tag{22}$$

In particular, if C is a loop that starts and ends at v, then

$$H_C x_v = x_v. \tag{23}$$

Thus a global section is determined by a base vector x_v that is fixed by every loop holonomy.

Conversely, suppose $z \in \mathbb{R}^2$ satisfies $H_C z = z$ for every loop C based at v. Choose one path p_u from v to each node u and define

$$x_u = T_{p_u} z. \tag{24}$$

This definition does not depend on the chosen path. Indeed, if q_u is a second path from v to u, then following p_u and returning along q_u forms a loop at v. Hence

$$T_{q_u}^{-1} T_{p_u} z = z \quad \Longrightarrow \quad T_{p_u} z = T_{q_u} z. \tag{25}$$

Extending a path to w by the edge $w \to u$ now gives $x_u = T_{u \leftarrow w} x_w$, so the vectors in Eq. (24) form a global section. We have proved

$$\Gamma(\mathcal{F}) \cong \{ z \in \mathbb{R}^2 \mid H_C z = z \text{ for every loop } C \text{ based at } v \}. \tag{26}$$

If the connection is flat, every H_C equals I_2, so every $z \in \mathbb{R}^2$ defines a global section. Conversely, suppose one loop has $H_C = R(\theta)$ with $\theta \not\equiv 0 \pmod{2\pi}$. Then

$$\det(R(\theta) - I_2) = 2 - 2\cos\theta = 4\sin^2(\theta/2) > 0, \tag{27}$$

so $\ker(R(\theta) - I_2) = \{0\}$. The intersection in Eq. (26), and therefore $\Gamma(\mathcal{F})$, is then $\{0\}$. ■

Appendix C. Experimental Design and Model Comparisons

This section collects the reproducibility details summarized in Section 5. It first defines the graph distributions and targets. It then gives the common model and evaluation protocol, followed by the separately trained baselines used to test alternative predictors of the target.

C.1. Graph Distributions and Targets

High-homophily graph family. The main distribution $\mathcal{D}_{\mathrm{GU}}^{\mathrm{hi}}$ uses high-homophily GraphUniverse graphs generated from a degree-corrected stochastic block model (Van Langendonck et al., 2026). Table 2 defines the universe and graph-level parameters.

Table 2: **High-homophily GraphUniverse specification.** Graph-level quantities are sampled independently within the stated ranges.

Symbol	Definition	Value or range
$s_{\mathcal{U}}$	Universe seed	42
$K_{\mathcal{U}}$	Universe communities	20
B	Edge-propensity matrix	$B \in \mathbb{R}^{20\times 20}$, variance 1.0
p	Node-feature dimension	15
$\mathrm{Var}(m_k)$	Community-center variance	0.2
$\mathrm{Var}(x_v \mid c_v = k)$	Within-community variance	0.4
$n = \lvert V\rvert$	Nodes per graph	$[100, 160]$
K_G	Communities represented in G	$\{5, 6, 7\} \subseteq \{1, \ldots, K_{\mathcal{U}}\}$
$\bar{d}$	Mean degree	$[5.0, 7.0]$
γ	Power-law exponent	$[2.0, 2.5]$
δ_{deg}	Degree-separation parameter	$[0.5, 1.0]$
h	Homophily parameter	$[0.9, 1.0]$

Edge-generation model. Let $B \in \mathbb{R}^{20\times 20}$ denote the fixed universe-level edge-propensity matrix. Conditional on community labels c_u, c_v and positive Pareto degree factors θ_u, θ_v, the edge probability for nodes u and v is proportional to $B_{c_u c_v}\theta_u\theta_v$ and clipped to $[0, 1]$. The generator makes each graph undirected and connected. Data generation and model initialization vary independently across seeds.

Triangle target. The triangle target is the training-standardized log count defined in Section 5. Test graphs never contribute to μ_s or σ_s.

Low-homophily dataset variant. The low-homophily dataset variant retains triangle counting, the graph-size and degree ranges in Table 2, the test set $\mathcal{D}_{\mathrm{test},8}^{(s)}$, and $s \in \mathcal{S}$. It changes the feature and community-homophily configuration and tests whether the twist–error result is unique to the high-homophily distribution. The frozen-intervention conclusions do not use this dataset variant.

Fixed-degree control. The fixed-degree control uses random 6-regular graphs with 100 nodes and 300 edges. If s is the data seed and i indexes a graph within that seed, then

$$G_i^{(s)} \sim \texttt{random_regular_graph}(6, 100, \texttt{seed} = 1000s + i). \tag{28}$$

Graph size and every node degree are constant, while wiring and triangle count vary. Independent 15-dimensional Gaussian features break node symmetry. Constant all-ones features test anonymous message passing.

Community-detection control. Community detection uses the high-homophily graph family and predicts, for every node, one of the $K_{\mathcal{U}} = 20$ universe-level community labels. The task comparison uses $N_{\text{train}} = 30$, $N_{\text{test}} = 8$, and $s \in \mathcal{S}$. It is a loop-nonspecific contrast because clustered features and homophilous neighborhoods provide direct predictive cues without requiring closed paths.

C.2. Models, Optimization, and Evaluation

Tables 3 and 4 define the complete configuration. Model equations are given in Appendix A.

Table 3: **Hyperparameter configurations of the trained neural models.**

Hyperparameter	NSD	NSP	ISN	SCCN
Propagation layers	3	3	3	3
Hidden channels	32	32	32	32
Stalk dimension d	2	2	2	—
Connection across layers	Exact-tied	Exact-tied	Fixed identity	—
Pooling	Mean	Mean	Mean	Mean
Readout	MLP	MLP	MLP	MLP
Input projection activation	ELU	ELU	ELU	ELU
Sheaf Learner	Linear + tanh	Linear + tanh	Fixed identity	—
Diffusion nonlinearity σ	ELU	ELU	ELU	—
Activation-magnitude parameter ϵ	tanh-parameterized	tanh-parameterized	tanh-parameterized	—
Optimizer	Adam	Adam	Adam	Adam
Learning rate	10^{-2}	10^{-2}	10^{-2}	10^{-2}
Batch size	8	8	8	8
Input Dropout	0.1	0.1	0.1	0.1
Hidden Dropout	0.1	0.1	0.1	0.1
Weight Decay	0	0	0	0
Parameter initialization	Kaiming-uniform	Kaiming-uniform	Kaiming-uniform	Kaiming-uniform
Objective	MSE	MSE	MSE	MSE
Maximum epochs	100	100	100	100
Early Stopping	15	15	15	15
Checkpoint selection	Min. $\mathcal{D}_{\text{val}}^{(s)}$ MSE	Min. $\mathcal{D}_{\text{val}}^{(s)}$ MSE	Min. $\mathcal{D}_{\text{val}}^{(s)}$ MSE	Min. $\mathcal{D}_{\text{val}}^{(s)}$ MSE
Model seeds	10 (0–9)	10 (0–9)	10 (0–9)	10 (0–9)

Sheaf Learner and nonlinearities. The Sheaf Learner applies a linear map to edge features followed by tanh. The input feature projection uses ELU, and the diffusion nonlinearity $\sigma = \text{ELU}$ is applied to $L_{\mathcal{F}}XW$ in the NSD update. Following the terminology of Bodnar et al. (2022), the additional activation-magnitude parameter $\epsilon \in [-1, 1]^d$ controls the coefficient $(1 + \epsilon)$ on the residual features. In our implementation it is obtained by passing a learned parameter directly through tanh. Thus tanh is used for restriction maps

and ϵ, whereas ELU is used for projected and diffused features. These component-level details apply to the sheaf architectures; SCCN does not use a Sheaf Learner.

Table 4: **Data and evaluation sets.** All four sets are mutually disjoint. Neither test set participates in checkpoint selection.

Set	Cardinality	Generator seed
$\mathcal{D}^{(s)}_{\text{train}}$	$N_{\text{train}} \in \{30, 100, 300, 1000, 3000\}$	$100 + s$
$\mathcal{D}^{(s)}_{\text{val}}$	$N_{\text{val}} = 64$	$6000 + s$
$\mathcal{D}^{(s)}_{\text{test},8}$	$N_{\text{test}} = 8$	$900 + s$
$\mathcal{D}_{\text{test},200}$	$N_{\text{test}} = 200$	7777

Test-set-size robustness. The same frozen checkpoint is evaluated on both $\mathcal{D}^{(s)}_{\text{test},8}$ and $\mathcal{D}_{\text{test},200}$. There is no retraining or checkpoint reselection. This comparison changes the number and sample of evaluation graphs, not the fitted model. The 200 raw graphs are identical for every s, while their standardized targets use the training statistics (μ_s, σ_s) from $\mathcal{D}^{(s)}_{\text{train}}$.

C.3. Training-Time Baselines and Positive Controls

Shortcut baselines. The training-mean predictor always outputs $z_s = 0$. Ridge regression receives node count, edge count, and the minimum, maximum, mean, standard deviation, and median of node degree. It never receives the adjacency matrix or enumerated triangles. Its success on GraphUniverse exposes graph-size and degree shortcuts but does not show that the neural models use the same computation.

Identity-trained baseline. The Identity Sheaf Network (ISN) fixes every transport to I_2 throughout training. It differs from post-training Identity replacement, which changes the connection inside a checkpoint trained with nonidentity transports.

Explicit-triangle positive control. The three-layer Simplicial Complex Convolutional Network (SCCN) operates on the two-dimensional clique complex and mean-pools rank-two simplex features (Yang et al., 2022). The clique lift explicitly enumerates the target triangles, so SCCN is a privileged positive control rather than a baseline with the same node-and-edge representation.

Appendix D. Constructing and Characterizing the Connection Replacements

This section defines the three frozen-checkpoint replacements from Section 4.2. It first constructs the synchronized-flat connection and then specifies the Shuffled and Identity references. It also describes what the replacements change and reports the numerical exactness checks. The spectral diagnostic appears separately in Section H. The NSP reliance results and the depth-and-tying control are reported in Section E.1.

D.1. Synchronized Flat Construction and Exact Flatness

The construction identifies planar rotations with unit complex numbers. Let i denote the imaginary unit, $i^2 = -1$, and define

$$R(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}, \qquad e^{i\alpha} = \cos\alpha + i\sin\alpha. \tag{29}$$

For a connected component $G_c = (V_c, E_c)$, choose an oriented-edge set $E_c^+ \subset V_c \times V_c$ containing exactly one of (u, v) and (v, u) for every undirected edge $\{u, v\} \in E_c$. The ordered pair (u, v) denotes transport from v to u. For each $(u, v) \in E_c^+$, encode the learned transport by

$$T_{u\leftarrow v} = R(\alpha_{uv}) \quad \longleftrightarrow \quad z_{uv} = e^{i\alpha_{uv}} = (T_{u\leftarrow v})_{11} + i(T_{u\leftarrow v})_{21}. \tag{30}$$

Thus z_{uv} is the phase attached to the directed edge $v \to u$, whereas the node phases introduced below are denoted by ϕ_u. The reverse transport is $T_{v\leftarrow u} = T_{u\leftarrow v}^{-1} = R(-\alpha_{uv})$ and has complex phase $\overline{z_{uv}}$.

The learned transports are inserted into the Hermitian phase-adjacency matrix $\mathcal{H}_c \in \mathbb{C}^{|V_c|\times|V_c|}$ as follows:

$$(\mathcal{H}_c)_{uv} = \begin{cases} z_{uv}, & (u, v) \in E_c^+, \\ \overline{z_{vu}}, & (v, u) \in E_c^+, \\ 0, & \text{otherwise.} \end{cases} \tag{31}$$

In words, a stored transport $T_{u\leftarrow v}$ places z_{uv} in row u, column v, and places its conjugate in row v, column u. Diagonal and nonedge entries are zero. Consequently, $(\mathcal{H}_c)_{vu} = \overline{(\mathcal{H}_c)_{uv}}$, so $\mathcal{H}_c^* = \mathcal{H}_c$. The matrix has the sparsity pattern of the graph adjacency matrix, but each nonzero entry records an oriented learned rotation rather than a scalar edge weight.

The role of $\mathcal{H}_c$ becomes clear in the flat case. A flat connection can be written using one angle ϕ_u per node, with

$$z_{uv} = e^{i(\phi_u - \phi_v)}. \tag{32}$$

If D_c is the degree matrix, this factorization implies

$$\mathcal{H}_c\big(e^{i\phi_u}\big)_{u\in V_c} = D_c\big(e^{i\phi_u}\big)_{u\in V_c}. \tag{33}$$

Equivalently, the normalized matrix

$$\widehat{\mathcal{H}}_c = D_c^{-1/2}\mathcal{H}_c D_c^{-1/2} \tag{34}$$

has eigenvalue one with eigenvector $D_c^{1/2}(e^{i\phi_u})_{u\in V_c}$. A learned connection is generally not flat, so this relation holds only approximately. Angular synchronization uses the leading eigenvector q_c of $\widehat{\mathcal{H}}_c$ as a spectral relaxation and estimates each node angle by $\phi_u = \mathrm{Arg}((q_c)_u)$, where $\mathrm{Arg} : \mathbb{C}\setminus\{0\} \to (-\pi, \pi]$ is the principal argument. Multiplying q_c by a global phase $e^{i\beta}$ adds the same β to every ϕ_u, so all differences $\phi_u - \phi_v$ remain unchanged. If $|(q_c)_u| \le 10^{-12}$, the implementation uses $\phi_u = 0$. The production runs never triggered this fallback.

Algorithm 1. Synchronized-flat reconstruction on one graph component

Input: A connected component $G_c = (V_c, E_c)$ and learned transports $T_{u\leftarrow v} \in \mathrm{SO}(2)$.
Output: Replacement transports $T^{\mathrm{flat}}_{u\leftarrow v}$ with identity holonomy on every cycle.

Encode edge rotations

Choose $E_c^+ \subset V_c \times V_c$.
$\mathcal{H}_c \leftarrow 0 \in \mathbb{C}^{|V_c|\times|V_c|}$.; `// start with an empty constraint matrix`
for $(u, v) \in E_c^+$ **do**
  $z_{uv} \leftarrow (T_{u\leftarrow v})_{11} + i(T_{u\leftarrow v})_{21}$.; `// encode the rotation as one complex number`
  $(\mathcal{H}_c)_{uv} \leftarrow z_{uv}$ and $(\mathcal{H}_c)_{vu} \leftarrow \overline{z_{uv}}$.
$D_c \leftarrow \mathrm{diag}(\deg_{G_c}(u) \mid u \in V_c)$.
$\widehat{\mathcal{H}}_c \leftarrow D_c^{-1/2}\mathcal{H}_c D_c^{-1/2}$.

Estimate node angles

Choose $q_c \in \arg\max_{\|q\|_2=1} q^*\widehat{\mathcal{H}}_c q$.; `// best global agreement with the edge rotations`
for $u \in V_c$ **do**
  if $|(q_c)_u| > 10^{-12}$ **then**
    $\phi_u \leftarrow \mathrm{Arg}((q_c)_u)$.; `// read node u's angle from the global fit`
  else
    $\phi_u \leftarrow 0$.; `// fallback if an angle is numerically undefined`

Construct the flat connection

for $(u, v) \in E_c^+$ **do**
  $T^{\mathrm{flat}}_{u\leftarrow v} \leftarrow R(\phi_u - \phi_v)$.; `// edge rotation = endpoint-angle difference`
  $T^{\mathrm{flat}}_{v\leftarrow u} \leftarrow (T^{\mathrm{flat}}_{u\leftarrow v})^{-1}$.
return $T^{\mathrm{flat}} = \{T^{\mathrm{flat}}_{u\leftarrow v}, T^{\mathrm{flat}}_{v\leftarrow u} \mid (u, v) \in E_c^+\}$.

For an exact-tied checkpoint, synchronization is run once per graph component and the component-wise outputs are assembled into the graph-wide replacement connection. The Cayley-parameterized implementation wraps $\phi_u - \phi_v$ to the principal representative $\mathrm{Arg}(e^{i(\phi_u-\phi_v)})$, converts that angle back to the model's endpoint parameters, rebuilds the normalized sheaf Laplacian, and reuses it in all three propagation layers while evaluating the frozen checkpoint.

For any cycle $C = (v_0, \ldots, v_{k-1}, v_0)$, the replacement phases satisfy

$$\prod_{j=0}^{k-1} e^{i(\phi_{v_{j+1}} - \phi_{v_j})} = e^{i\sum_{j=0}^{k-1}(\phi_{v_{j+1}} - \phi_{v_j})} = 1, \qquad v_k = v_0. \tag{35}$$

Thus every cycle holonomy is identity. This replacement is a pure-gauge connection, not a gauge transformation of the learned connection. A gauge transformation would preserve holonomy up to conjugation. The construction preserves the graph, edge set, architecture, non-connection parameters, and edge-coupled $\mathrm{SO}(2)$ propagation while replacing all nonidentity cycle products.

D.2. Shuffled and Identity Reference Replacements

Shuffled. Choose one orientation E^+ of the undirected edges and sample a permutation $\pi : E^+ \to E^+$ within each graph. If $T_e = R(\alpha_e)$ for $e \in E^+$ and $\bar{e}$ denotes the reverse orientation, Shuffled sets

$$T_e^{\mathrm{shuffle}} = R(\alpha_{\pi(e)}), \qquad T_{\bar{e}}^{\mathrm{shuffle}} = (T_e^{\mathrm{shuffle}})^{-1}. \tag{36}$$

This preserves the multiset of learned angles and inverse consistency. It changes which edge receives each angle and thereby changes the loop products.

Identity. Identity sets

$$T_e^{\mathrm{id}} = I_2 \tag{37}$$

for both orientations of every edge. Both replacements reuse the resulting connection across all three layers and rebuild the normalized sheaf Laplacian. No other checkpoint parameter changes.

D.3. What the Replacements Change

Displacement measures. The replacements in Table 1 alter local edge angles and loop products in different proportions. Figure 6 compares mean edge-angle change, twist remaining after intervention, and twist removed from the learned connection. None of the replacements holds local transport fixed while changing only selected cycle products.

Within-size associations. Within training sizes, the amount of twist removed by Identity is associated with the MSE increase under Identity. Pearson $r = 0.461$ with Holm-adjusted $p = 0.0053$, and Spearman $\rho = 0.513$ with Holm-adjusted $p = 0.0010$. Identity also deletes all local nonidentity transport. The association therefore indicates sensitivity to the full learned connection, not holonomy-specific causality.

D.4. Numerical Exactness Checks

Tolerance checks. Before running the experimental sweeps, we set a numerical tolerance of 2×10^{-5} for the measurement and intervention checks. Table 5 reports the maximum observed error. Every nonzero value is more than an order of magnitude below that tolerance.

Table 5: **Numerical validation of measurement and intervention code.** The maximum error is taken over the corresponding validation cases.

Check	Maximum error
Basis invariance of retained loop readouts	5.96×10^{-8}
SO(2) loop determinant relative to one	1.19×10^{-7}
Reverse rotation versus matrix inverse	5.96×10^{-8}
Reconstruction of requested rotations by the model parameterization	5.96×10^{-8}
Identity readouts	0
Synchronized-flat loop product	2.38×10^{-7}
Shuffled-angle multiset preservation	0

Appendix E. Comparison of NSP and NSD

This section compares NSP and NSD under the same empirical questions. Where both results are available, each subsection reports NSP first and NSD second so that the architecture comparison is easy to scan. When a result already appears in the main text or an earlier appendix section, we cite that result rather than repeat it.

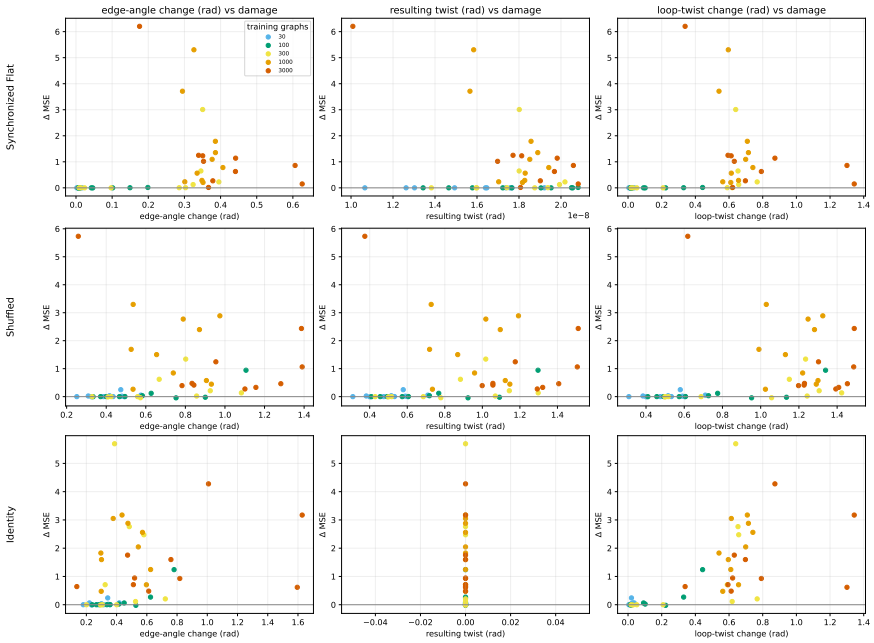


Figure 6: **Transport and loop displacement do not give a common explanation for intervention-induced MSE increases.** The interventions alter edge angles and loop products in different ways. Within-size analyses do not show that the MSE increases under Synchronized Flat or Shuffled are predicted by the measured edge and loop displacement.

E.1. Connection Replacement Sensitivity

NSP. Figure 7 reports the ten-seed NSP sweep. It shows the learned-checkpoint error and the within-checkpoint MSE increase after each connection replacement. At $N_{\text{train}} = 3000$, every NSP checkpoint has an MSE increase under Synchronized Flat, Shuffled, and Identity.

NSD. The main intervention curves appear in Figure 3b. Here we add checkpoint-level transport and loop readouts together with within-checkpoint inference. Define the learned-to-Flat edge displacement of a graph as the mean absolute angular difference between corresponding learned and synchronized-flat edge rotations. Figure 8 relates this displacement, triangle twist, and the MSE increase under Synchronized Flat across all 50 exact-tied NSD checkpoints.

Within-checkpoint inference. Table 6 gives the inferential summary. The confidence intervals are two-sided t intervals for the mean MSE difference. The paired test is an exact two-sided sign-flip randomization test on the ten checkpoint-level differences, with Holm correction across all 15 intervention-by-size tests.

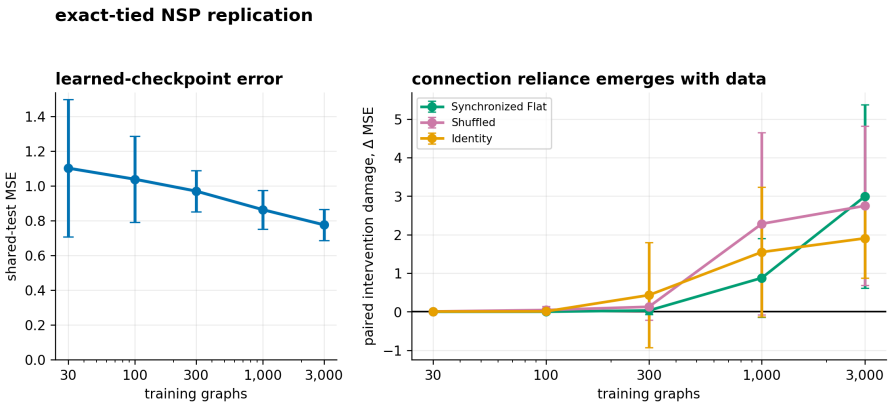


Figure 7: **Exact-tied NSP becomes sensitive to connection interventions after training on more graphs.** The left panel shows Learned MSE. The right panel shows the intervention-induced MSE increase relative to Learned. Points are means and error bars are sample standard deviations over ten checkpoints evaluated on $\mathcal{D}_{\text{test},200}$.

Table 6: **NSD intervention inference on $\mathcal{D}_{\text{test},200}$.** The reported difference is intervention MSE minus Learned MSE. Values are mean ± sample standard deviation (SD) over ten checkpoints. The confidence interval (CI) is a two-sided 95% t interval. The sign column counts checkpoints with an MSE increase. The final column reports the exact two-sided sign-flip p-value after Holm correction across all 15 tests.

N_{train}	Intervention	Δ, mean ± SD	95% CI	$\Delta > 0$	Holm p
30	Synchronized Flat	0.000 ± 0.000	$[-0.000, 0.000]$	6/10	0.3750
30	Shuffled	0.037 ± 0.077	$[-0.018, 0.092]$	8/10	0.2871
30	Identity	0.041 ± 0.075	$[-0.012, 0.095]$	7/10	0.2930
100	Synchronized Flat	0.001 ± 0.005	$[-0.002, 0.005]$	6/10	0.6445
100	Shuffled	0.104 ± 0.298	$[-0.110, 0.317]$	7/10	0.6445
100	Identity	0.158 ± 0.390	$[-0.121, 0.437]$	8/10	0.3613
300	Synchronized Flat	0.402 ± 0.938	$[-0.269, 1.073]$	7/10	0.2109
300	Shuffled	0.228 ± 0.439	$[-0.086, 0.542]$	5/10	0.3750
300	Identity	1.195 ± 1.900	$[-0.164, 2.554]$	6/10	0.2500
1,000	Synchronized Flat	1.532 ± 1.692	$[0.321, 2.742]$	10/10	0.0293
1,000	Shuffled	1.668 ± 1.118	$[0.868, 2.468]$	10/10	0.0293
1,000	Identity	1.956 ± 0.961	$[1.268, 2.644]$	10/10	0.0293
3,000	Synchronized Flat	1.276 ± 1.790	$[-0.004, 2.557]$	10/10	0.0293
3,000	Shuffled	1.282 ± 1.699	$[0.067, 2.498]$	10/10	0.0293
3,000	Identity	1.512 ± 1.265	$[0.608, 2.417]$	10/10	0.0293

Interpretation. At $N_{\text{train}} = 1000$ and $N_{\text{train}} = 3000$, every checkpoint worsens under every intervention, and all six Holm-adjusted sign-flip tests give $p = 0.0293$. The t interval for Synchronized Flat at $N_{\text{train}} = 3000$ narrowly includes zero because the MSE difference has a large across-seed SD. The sign-flip test instead detects the unanimous direction of the within-checkpoint effects. The sweep shows checkpoint reliance after training on 1,000 or 3,000 graphs. It does not show a stable ranking of the three interventions or a within-size relation between twist magnitude and the MSE increase under Synchronized Flat.

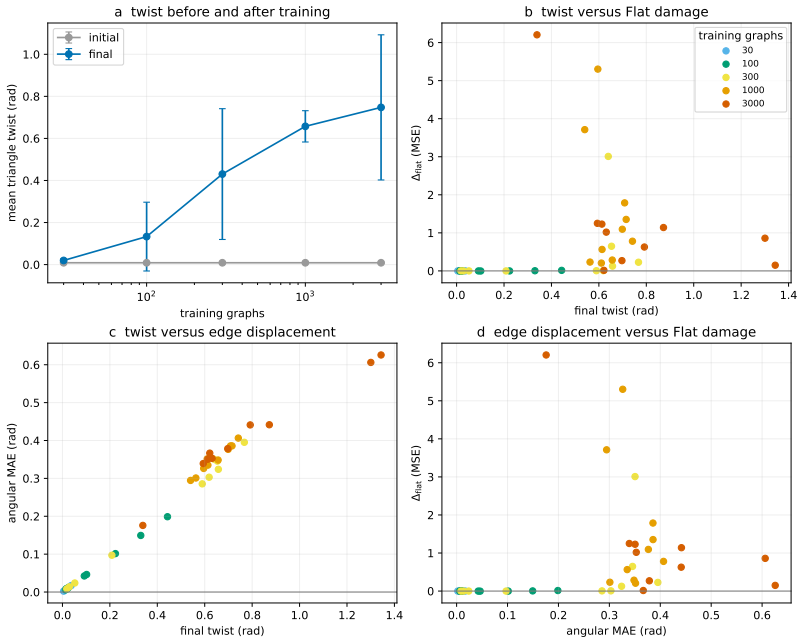


Figure 8: **NSD triangle twist changes with training size, but twist does not explain the MSE increase under Synchronized Flat within size.** Panel (a) shows mean initial and final twist with sample-standard-deviation bars over ten seeds. Panels (b–d) show individual checkpoints. After within-size centering, twist and the MSE increase under Synchronized Flat have Pearson $r = -0.190$ with $p = 0.185$ and Spearman $\rho = 0.077$ with $p = 0.593$.

The NSP and NSD sweeps therefore agree that frozen checkpoints become sensitive to connection replacement after training on more graphs. Their training-time learned-versus-identity comparisons differ, as reported in Section F.

E.2. Depth and Tying Control

NSD. Only the corrected ten-seed NSD comparison is reported here. On $\mathcal{D}_{\text{test},200}$ at $N_{\text{train}} = 3000$, ordinary NSD, exact-tied NSD, and ISN-NSD obtain 0.576 ± 0.104, 0.583 ± 0.093, and 0.636 ± 0.134 MSE on GraphUniverse, respectively, over ten seeds. The training-mean MSE is 0.969. On random 6-regular graphs, the corresponding values are 1.035 ± 0.031, 1.035 ± 0.030, and 1.034 ± 0.029, against a training mean of 1.026. Thus tying does not materially degrade NSD on GraphUniverse, while none of these variants gains a counting advantage on fixed-degree graphs.

E.3. Task Comparison across Restriction Families

Here twist change means trained minus initial triangle-weighted mean twist, as in Section 4.1.

NSP. At $N_{\text{train}} = 30$, the ten-seed mean SO(2) twist change is 0.0134 ± 0.0139 on triangle counting and 0.0317 ± 0.0118 on community detection.

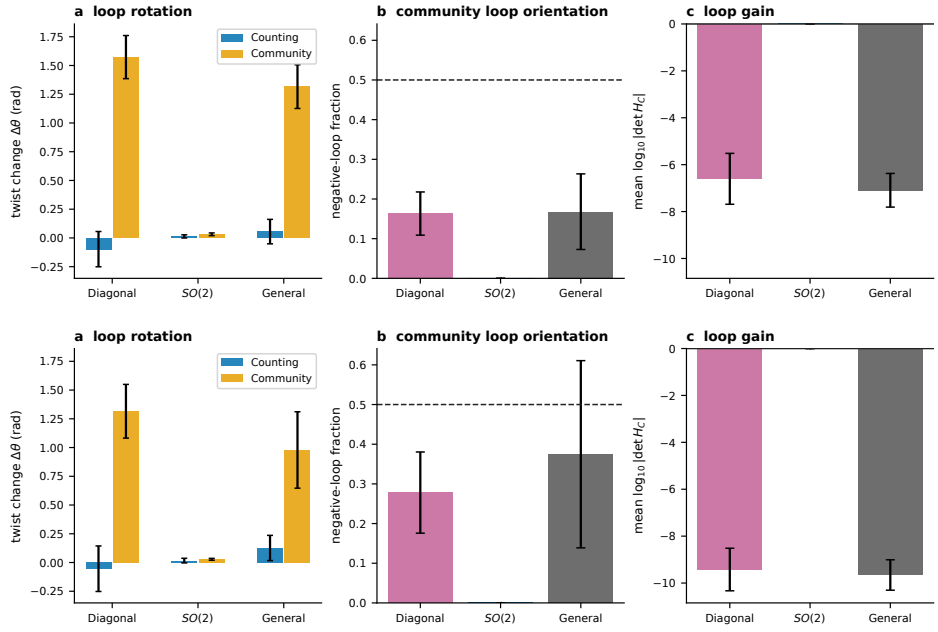


Figure 9: **Community detection produces more loop rotation than triangle counting.** The upper row reports exact-tied NSP and the lower row exact-tied NSD at $N_{\text{train}} = 30$. Within each row, panel (a) compares twist change across triangle counting and community detection, panel (b) reports the negative-loop fraction for community detection, and panel (c) reports mean log absolute determinant. Bars show means and sample standard deviations over ten seeds. For SO(2), gain is one and flip is +1, so only twist varies. Diagonal and general loop products are shown for comparison but are not strict holonomy.

NSD. The corresponding NSD values are 0.0166 ± 0.0202 and 0.0279 ± 0.0082. Thus, for both architectures, community detection produces more twist than triangle counting. The comparison provides no evidence that triangle counting uniquely recruits holonomy.

Figure 9 reports the full restriction-family comparison for both architectures.

E.4. Twist and Counting Error

Recall that $\Delta_{\text{const}} = \mathcal{L}_{\text{model}} - \mathcal{L}_{\text{mean}}$ compares a checkpoint's MSE on $\mathcal{D}^{(s)}_{\text{test},8}$ with the training-mean predictor on the same graphs. Lower values are better.

NSP. Figure 10 gives Pearson $r = 0.41$ with $p = 0.07$ across the 20 NSP checkpoints.

NSD. The corresponding 20-checkpoint NSD result already appears in Figure 4a. It gives Pearson $r = 0.33$ with $p = 0.16$.

Combined comparison. Combining the 20 checkpoints from each architecture gives Pearson $r = 0.303$, $p = 0.0572$, and Spearman $\rho = 0.180$, $p = 0.2651$. Neither architecture shows the negative association expected if greater twist improved counting. We do not interpret the weak positive trends as evidence that twist is generally harmful.

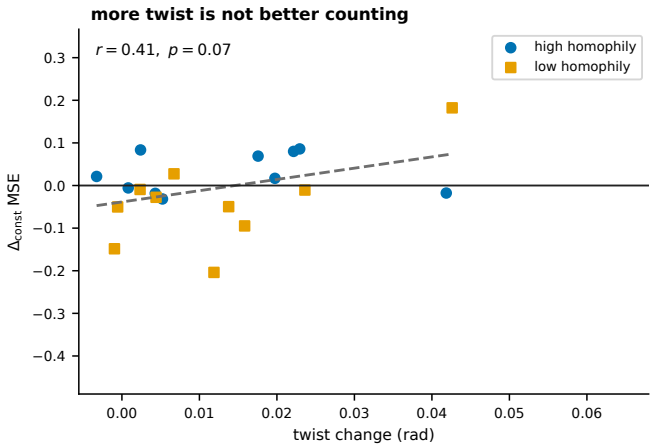


Figure 10: **More NSP twist is not associated with lower counting error.** Points are exact-tied NSP checkpoints trained on the high- and low-homophily counting datasets, with ten seeds for each dataset. The vertical axis is MSE relative to the training-mean predictor, so lower values are better. The positive Pearson correlation, $r = 0.41$ with $p = 0.07$, is opposite to the negative association expected if greater twist improved counting.

E.5. Fixed-Degree Controls

NSP. Figure 11 reports the exact-tied NSP controls. At $N_{\text{train}} = 30$, diagonal, SO(2), and general NSP have $\Delta_{\text{const}} = 0.005 \pm 0.073$, 0.010 ± 0.067, and -0.004 ± 0.059, respectively, over ten seeds. In the separate three-seed size sweep, the mean SO(2) twist changes at $N_{\text{train}} = 30, 100, 300, 1000, 3000$ are 0.007 ± 0.011, 0.024 ± 0.033, 0.293 ± 0.124, 0.366 ± 0.106, and 0.448 ± 0.260 radians. The model remains near the training-mean predictor while twist increases, although the three-seed curve is descriptive.

NSD. Figure 12 reports the corrected fixed-degree NSD control. At $N_{\text{train}} = 30$, diagonal and SO(2) NSD have $\Delta_{\text{const}} = 0.058 \pm 0.099$ and 0.055 ± 0.076, respectively. The SO(2) twist means at $N_{\text{train}} = 30, 100, 300, 1000, 3000$ are 0.045 ± 0.043, 0.071 ± 0.085, 0.255 ± 0.193, 0.127 ± 0.110, and 0.071 ± 0.037 radians. Diagonal and SO(2) NSD remain near the training-mean predictor while SO(2) twist changes non-monotonically. With constant node features, every node begins with the same feature vector and SO(2) twist remains at zero.

Appendix F. Training-Time Baselines and Alternative Predictors

Identity-trained models. Figure 13 shows that identity-trained NSD improves sharply with data and has lower mean error than learned-connection NSD at moderate and large training sizes. Identity-trained NSP does not show the same improvement. These comparisons use $\mathcal{D}^{(s)}_{\text{test},8}$ and ten seeds.

Degree-summary shortcut. Figure 14 reports the shortcut baseline across the same five training sizes. Degree ridge remains below learned and diagonal NSD in mean MSE

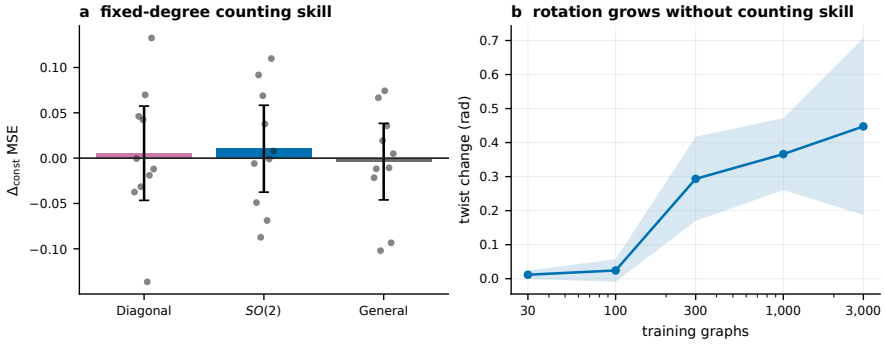


Figure 11: **Exact-tied NSP develops twist without improved counting on random 6-regular graphs.** Panel (a) compares diagonal, SO(2), and general NSP at $N_{\text{train}} = 30$ over ten seeds. Points are seeds and bars show means with sample-standard-deviation error bars. Panel (b) shows the SO(2) twist change across training sizes with sample-standard-deviation bands over three seeds.

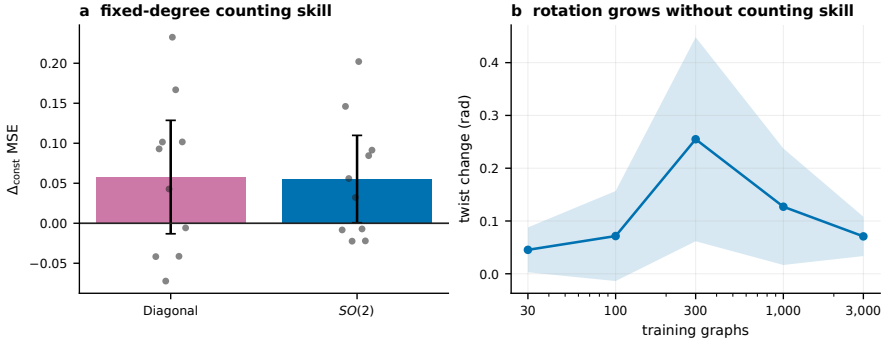


Figure 12: **Connection twist changes without improved counting on random 6-regular graphs.** Panel (a) compares diagonal and SO(2) NSD at $N_{\text{train}} = 30$. Points are seeds and bars show means with sample-standard-deviation error bars. Panel (b) shows mean SO(2) twist change across training sizes with sample-standard-deviation bands.

on $\mathcal{D}^{(s)}_{\text{test},8}$. Ridge receives graph size and degree summaries that are not directly preserved by mean pooling, so this comparison diagnoses information available in the sampled graph family rather than the computation used by NSD.

Explicit-triangle positive control. The three-layer SCCN reaches 0.077 ± 0.036 MSE at $N_{\text{train}} = 3000$ on $\mathcal{D}^{(s)}_{\text{test},8}$ over ten seeds. Its clique-complex input enumerates every triangle before learning. SCCN therefore confirms that the target is learnable when triangles are explicit. It does not provide a comparison with the same node-and-edge representation used by the sheaf networks.

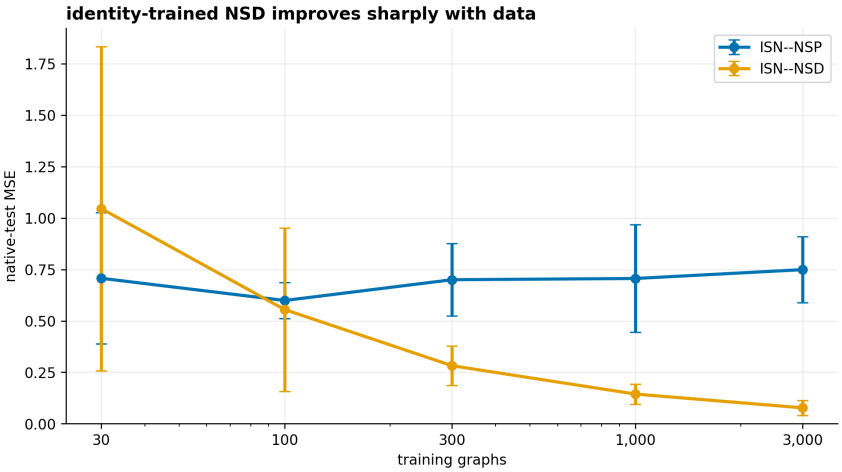


Figure 13: **Identity-trained NSD improves with data while identity-trained NSP does not.** Points show mean MSE on the seed-specific test sets, and error bars show sample standard deviations over ten seeds. This separately optimized comparison tests task-level value rather than frozen-checkpoint reliance.

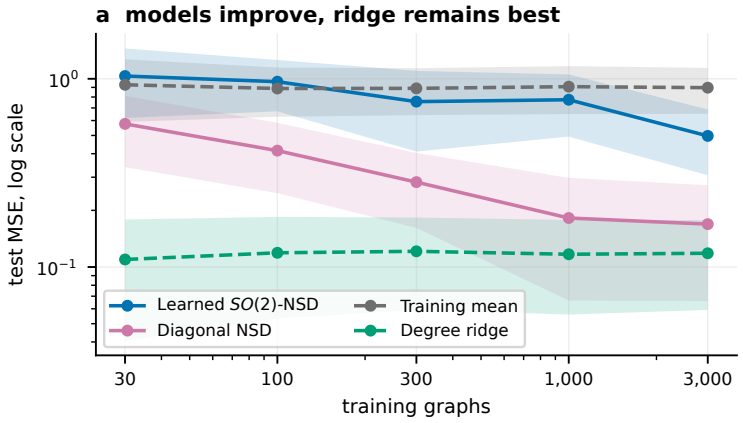


Figure 14: **Degree summaries provide a strong shortcut on GraphUniverse.** Degree ridge has lower mean MSE on $\mathcal{D}^{(s)}_{\text{test},8}$ than learned SO(2) NSD, diagonal NSD, and the training-mean predictor across all five training sizes. Curves show means and bands show sample standard deviations over ten seeds. Ridge receives node count, edge count, and five degree summaries but no adjacency matrix or enumerated triangles. Its performance shows that these graph summaries predict the target in this distribution. It does not show that NSD uses the same information.

Conclusion. Together, the ISN and SCCN controls show that frozen-checkpoint reliance does not make a learned nonidentity connection necessary for low counting error.

Appendix G. Limitations

The experiments use synthetic graphs, rank-two stalks, SO(2) transports, and triangle loops. Exact tying makes the same connection available for three-step composition but is only one architecture design. Results may differ for longer cycles, higher-rank bundles, nonorthogonal maps, or natural datasets.

Most experiments use ten trained seeds, but intervention effects remain highly variable and contain outliers. The five-size NSP fixed-degree sweep uses three seeds and is interpreted descriptively. The 200-graph intervention set reduces uncertainty from test sampling. It does not reduce variation caused by optimization or data generation.

Synchronized Flat is an analysis-time, graph-wide intervention. Although the reconstruction is exactly cycle-flat, it also changes individual edge transports, their assignment to edges, the connection adjacency, the normalized sheaf Laplacian, and later-layer representations. Moreover, the spectral relaxation does not guarantee the closest flat connection under an angular, chordal, or task-relevant distance. The increase in MSE therefore identifies sensitivity to this synchronized-flat replacement rather than to holonomy alone. Shuffled and Identity make still broader changes. An intervention with the same change magnitude that modifies selected loop products while preserving local message statistics would offer stronger causal identification.

Identity-trained NSD follows a separate optimization path. Its performance in this experiment shows that nonidentity transport is not necessary under the tested architecture and data. It does not show how the learned-connection checkpoint would adapt if fine-tuned after an intervention. Conversely, the frozen-checkpoint MSE increases show co-adaptation and reliance, not architectural necessity.

Appendix H. Spectral Diagnostic for Synchronized Flat

This diagnostic addresses two validation questions. It checks whether the synchronized replacement is globally cycle-flat and whether a global inconsistency measure agrees with the local triangle-twist readout. It does not test whether either quantity causes accurate counting.

Let $\lambda_1(\widehat{\mathcal{H}})$ be the largest eigenvalue of the normalized phase-adjacency matrix. We call $1 - \lambda_1(\widehat{\mathcal{H}})$ the *spectral frustration*. Equivalently, it is the smallest eigenvalue of the normalized connection Laplacian $I - \widehat{\mathcal{H}}$, where I is the identity matrix. For each checkpoint, graph-level quantities are averaged over the 200-graph test set $\mathcal{D}_{\text{test},200}$, and uncertainty bars show the sample standard deviation across ten trained seeds.

Figure 15 shows that Synchronized Flat restores $\lambda_1 = 1$ and zero spectral frustration up to numerical precision. Learned spectral frustration is strongly associated with triangle twist even after centering within training size (Pearson $r = 0.89$), so the local and global readouts track the same connection inconsistency. After the same within-size centering, spectral frustration does not predict the MSE increase under Synchronized Flat (Pearson $r = -0.09$, Spearman $\rho = 0.06$). Holm adjustment across the four correlation tests displayed in the figure gives adjusted p-values 4.36×10^{-28} and 5.91×10^{-18} for the pooled and within-size twist associations, and 1.0 for both within-size MSE-increase associations.

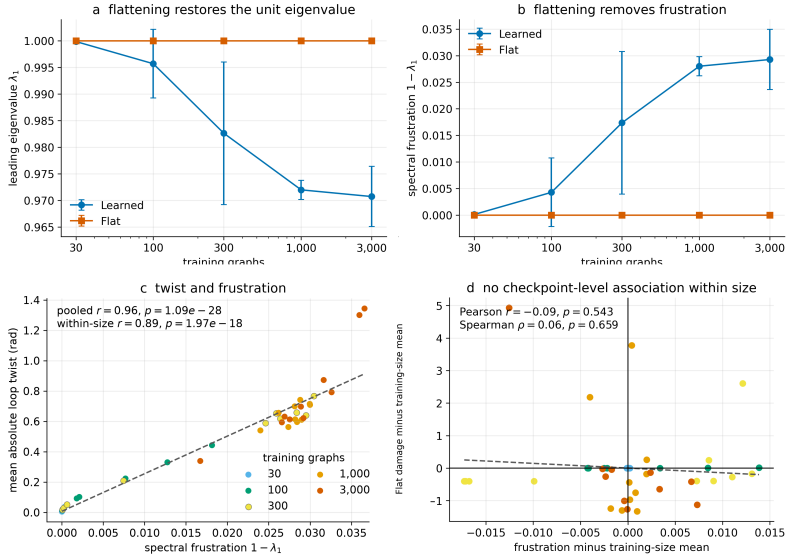


Figure 15: **Spectral diagnostics validate Synchronized Flat without identifying a counting mechanism.** Panels (a–b) compare the learned and reconstructed connections. Flattening restores the unit leading eigenvalue and removes spectral frustration across all five training sizes. Panel (c) shows that local triangle twist and global spectral frustration agree both pooled and within training size. Panel (d) centers frustration and the MSE increase under Synchronized Flat within each training size. Checkpoint-level deviations in frustration do not predict the MSE increase. Points in (c–d) are the 50 NSD checkpoints, with color denoting training size. Correlation p-values printed inside the panels are unadjusted. Holm-adjusted values are reported in the text. Spectral quantities are computed on $\mathcal{D}_{\text{test},200}$, and bars in (a–b) show sample standard deviations across ten seeds.